We can observe that the figures and its reflections are exactly the same distance from the line of reflection on both sides.

The line of reflection may be horizontal or vertical or slanting and also it may be on the shape or outside the shape.

Note

The line of reflection is the perpendicular bisector of the line joining at any point and its image

How to reflect a shape about a line?

B1G3N3

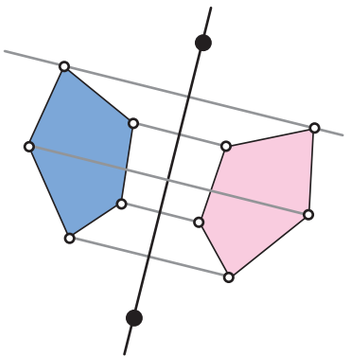


Fig. 4.11

To reflect the shape about the line of reflection, we have to reflect every vertex individually and then connect them again.

First, choose one of the vertices and draw the line through this vertex so that it is perpendicular to the line of reflection.

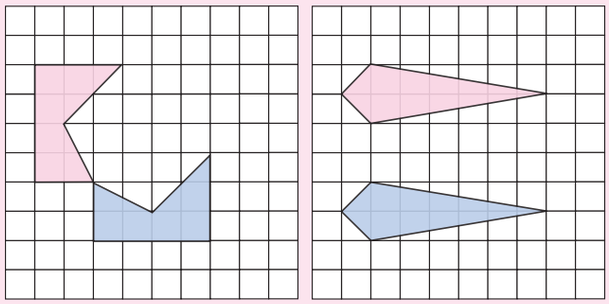
Now measure the distance from the vertex to the line of the reflection, and mark a point that has the same distance on the other side. It can be done by using either a ruler or a compass.

Repeat the process for all the other vertices of the shape.

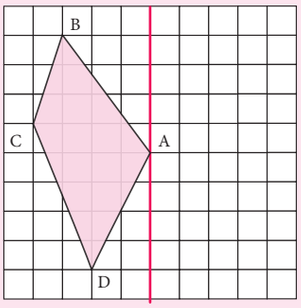
Finally connect all the reflected vertices in the correct order to get the reflection of the shape.

Try these

1. Draw the line of reflection in the following pictures.



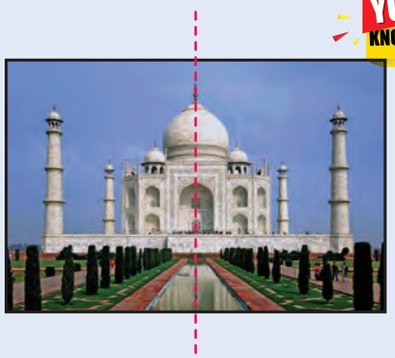
2. Reflect the shape with given line of reflection.



DO YOU KNOW?

Taj Mahal at Agra is planned by following the axis with Bilateral symmetrical design in plan and overall campus as the mirror image as shown in figure.

The symmetry in architecture is implied by its axiality or centrality in the form of the building. The monumental architecture often uses symmetrical design i.e. mirrored, which show stability, balance and control.



7th_Maths_Geometry_Chp4_New.indd 80 05-11-2019 17:35:53